Cloud Computing for ABC Enterprise

Key Contributions to Business Automation

CLOUD COMPUTING FUNDAMENTALS
ASSESSMENT 1

PROF DR. DIVYA LEEKHA

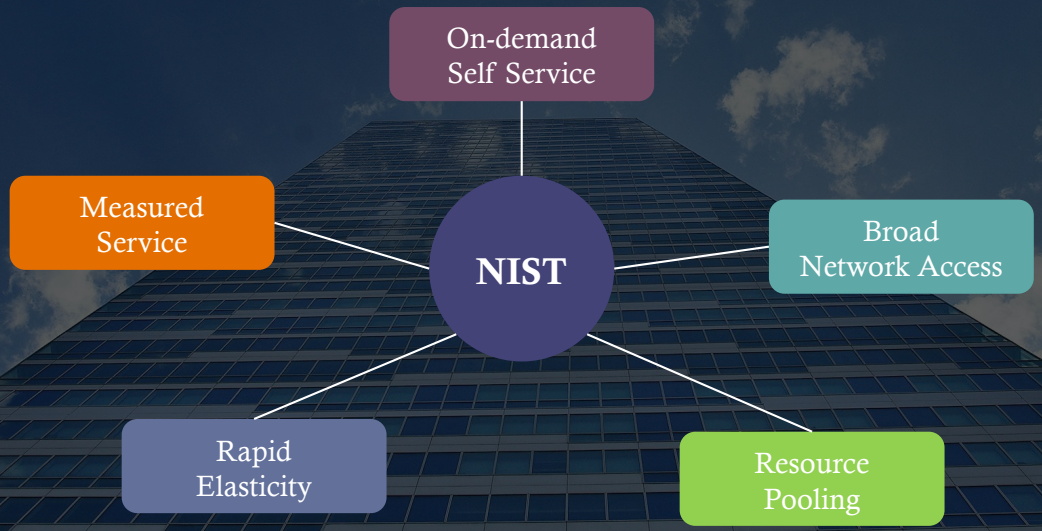
LUIS G. B. A. FARIA – ID A00187785

March, 2026

TORRENS UNIVERSITY AUSTRALIA

WHAT IS CLOUD COMPUTING?

On-demand delivery of compute, storage and services over the internet (Mell & Grance, 2011).



Key shift: from **owning capacity → renting capability**

TORRENS UNIVERSITY AUSTRALIA

BENEFITS OF CLOUD COMPUTING

B1 – Cost Efficiency

Traditional IT (CapEx)	Cloud – ABC (OpEx)
Large upfront hardware purchase	**No upfront servers**
Idle capacity during low demand	**Pay per compute hour.+ GB-month**
Full ops team required	**Provider manages infrastructure**
Fixed costs regardless of usage	**~80% reduction in start-up IT costs**

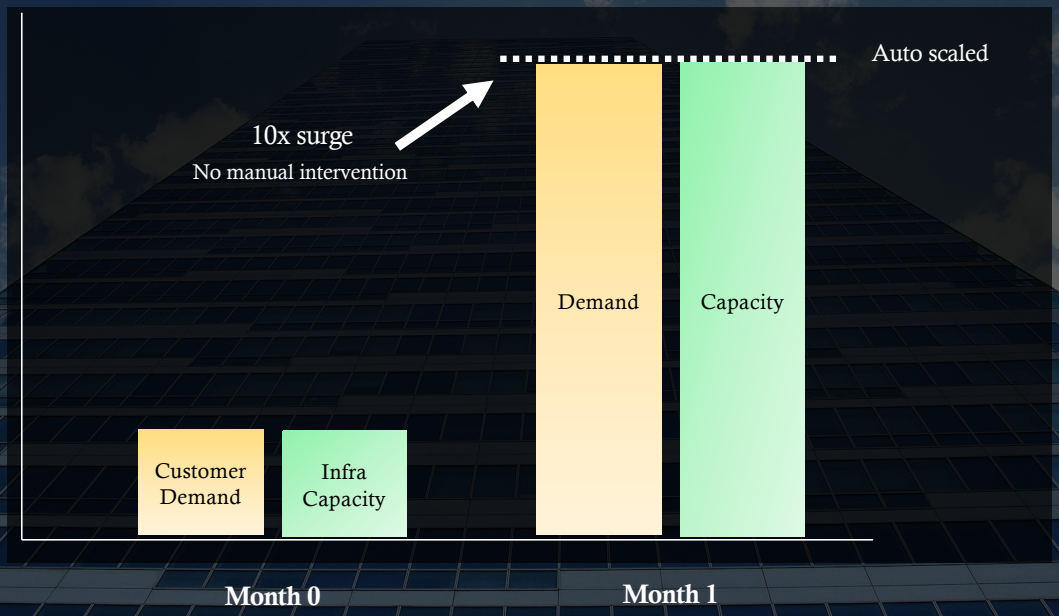
COST SHIFT

TORRENS UNIVERSITY AUSTRALIA

BENEFITS OF CLOUD COMPUTING

B2 & B3 – Scale without Limit & Less Overhead

Rapid Elasticity: ABC's 10x Growth absorbed by Auto Scaling



TORRENS UNIVERSITY AUSTRALIA

Challenges of Cloud Migration

Challenge	*Example*	*Mitigation*
Security & Privacy	PII + payment data	IAM least-privilege, MFA, encryption, shared responsibility model
Cost Volatility	Pay-as-you-go can spiral without guardrails	FinOps: budget alerts, resource tagging, rightsizing, reserved pricing
Vendor Lock-in + Skills Gap	Migration cost + mindset shift	Portability-first (containers, standard DBs), targeted cloud upskilling

TORRENS UNIVERSITY AUSTRALIA

Mitigation of Challenges

Phased migration approach, security best practices, compliance frameworks

Staff training programs, hybrid solutions for critical components

Recommended Models for ABC

Cloud Service Models

IaaS
Compute + Config
Flexibility

PaaS
Managed DB, load balancing,
serverless

SaaS
Limited customization

Cloud Deployment Models

Private cloud
High cost, limited elasticity

Hybrid cloud
Premature complexity

Public cloud
Scalable, automated,
cost-effective

TORRENS UNIVERSITY AUSTRALIA

Recommended Cost Models

Hybrid Cost Model

- **Reserved capacity:** stable tiers (web/app servers, databases)
- **Pay-as-you-go:** demand spikes via Auto Scaling
- **Spot instances:** background jobs + analytics pipelines

Conclusion

Summary of benefits for ABC

Next steps, final recommendations

TORRENS UNIVERSITY AUSTRALIA

REFERENCES

Alkhnbashi, O. S., Mohammad, R., & Hammoudeh, M. (2024). *Aspect-based sentiment analysis of patient feedback using large language models.* Big Data and Cognitive Computing, 8(12). https://doi.org/10.3390/bdcc8120167

Angelis, J. N., Murthy, R. S., Beaulieu, T., & Miller, J. C. (2024). *Better angry than afraid: the case of post data breach emotions on customer engagement.* IEEE Transactions on Engineering Management, 71, 2593–2605. https://doi.org/10.1109/TEM.2022.3189599

Bryman, A. (2016). *Social research methods* (5th ed.). Oxford University Press.

Cahya, V. N. C., Setyanto, R., & Paradise, P. (2025). *Analysis of the effectiveness of loyalty membership programs in increasing customer retention using net promoter score (NPS) with information system support*. Eduvest-Journal of Universal Studies, 5(9), 10974-10983.

Chen, E. (2023). *Growth product manager's handbook.* O'Reilly Media.

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Routledge.

Creswell, J. W., & Plano Clark, V. L. (2023). *Designing and conducting mixed methods research* (4th ed.). SAGE Publications.

Dawes, J. G. (2024). *The net promoter score: What should managers know? International Journal of Market Research*, 66(2–3), 182–198. https://doi.org/10.1177/14707853231195003

Dawes, J. G. (2024). N*et promoter and revenue growth: An examination across three industries.* Australasian Marketing Journal, 32(1), 4-18.

Field, A. (2018). *Discovering statistics using IBM SPSS Statistics* (5th ed.). SAGE Publications.

Godovykh, M., & Pizam, A. (2023). *Measuring patient experience in healthcare.* International Journal of Hospitality Management, 112, 103405. https://doi.org/10.1016/j.ijhm.2022.103405

Mar, J., & Armaly, P. (2023). *Mastering customer success.* O'Reilly Media.

Morgan, D. L. (2014). *Pragmatism as a paradigm for social research*. Qualitative Inquiry, 20(8), 1045–1053. https://doi.org/10.1177/1077800413513733

Pallant, J. (2020). *SPSS survival manual* (7th ed.). Routledge.

Polonsky, M. J., & Waller, D. S. (2019). Quantitative data analysis. *In Designing and managing a research project* (4th ed., pp. 222-254). SAGE Publications. https://doi.org/10.4135/9781544316499

Shankar, R., & Yip, A. (2024). *Transforming patient feedback into actionable insights through natural language processing: A knowledge discovery and action research study.* JMIR Formative Research. https://doi.org/10.2196/69699

Wohlin, C., Runeson, P., Höst, M., Ohlsson, M. C., Regnell, B., & Wesslén, A. (2012). *Experimentation in software engineering*. Springer.

Wohlin, C., & Runeson, P. (2021). Guiding the selection of research methodology in industry–academia collaboration in software engineering. *Information and Software Technology*, 140, 106678. https://doi.org/10.1016/j.infsof.2021.106678

Xiao, Y., Li, C., Thürer, M., Liu, Y., & Qu, T. (2022). *Towards lean automation: Fine-grained sentiment analysis for customer value identification.* Computers and Industrial Engineering, 169. https://doi.org/10.1016/j.cie.2022.108186

TORRENS UNIVERSITY AUSTRALIA

Thank you!

CLOUD COMPUTING FUNDAMENTALS
ASSESSMENT 1

PROF DR. DIVYA LEEKHA

LUIS G. B. A. FARIA – ID A00187785

March, 2026

TORRENS UNIVERSITY AUSTRALIA